DSC 4107 Homework 1

September 24, 2026

Rounding Requirements

1. Round all final numerical results to **four significant figures**.
2. Do not round intermediate results. Retain sufficient decimal places throughout the calculations and round only the final answer.

Calculator Policy

You are allowed to use the **Texas Instruments BA II Plus** financial calculator. A helpful tutorial can be found here: BA II Plus Tutorial (YouTube).
Important Requirements: If you choose to use the financial calculator, you must adhere to the following rules:

1. **Do NOT omit the mathematical formulas.** You must demonstrate your theoretical understanding by writing down the complete formula setup before providing the result.
2. **List the calculator parameters.** Alongside your formula, clearly state the inputs you entered into the calculator (e.g., $N = 6, I/Y = 5, \ldots, FV = 100$).

---

1 Question 1

Consider a corporate bond with a Face Value (Par Value) of \$1,000 and a maturity of 3 years. The current market requires a Yield to Maturity (YTM) of 8% for bonds with similar risk profiles. Please answer the following questions:

- Q1-1: Assume the bond has a fixed Coupon Rate of 10% and pays interest annually. Calculate the current market price of the bond.
- Q1-2: If this is a Zero-coupon bond (i.e., it pays no interest during the term and repays the face value at maturity), with all other conditions (3-year maturity, 8% YTM) remaining the same, what is its current price?
- Q1-3: Returning to the scenario in Q1-1 (10% coupon rate), if the bond pays interest quarterly instead of annually, calculate its theoretical price. Compare this price with the result from Q1-1 (state whether it is higher or lower).
- Q1-4: Assume the yield is quoted with Continuous Compounding (i.e., $r = 8\%$ continuously compounded). Using the cash flows from Q1-1 (10% annual coupon), calculate the theoretical price of the bond.
- Q1-5: Assume the bond's cash flows are no longer fixed but follow a Step-up Coupon structure: the coupon rate is 6% in Year 1, 8% in Year 2, and 12% in Year 3 (principal is repaid at the end of Year 3). Under discrete compounding (YTM = 8%), calculate the current price of this bond.
- Q1-6: Assume another 3-year corporate bond with a Face Value of \$1,000 and an annual coupon rate of 6% is currently trading at a market price of \$950. Calculate its Yield to Maturity (YTM). (Hint: you could use Excel or a financial calculator such as BA II Plus to solve for the rate).

2 Question 2

Scenario:
Consider a **3-year** Treasury bond with a **Face Value of $100** and a **Coupon Rate of 6%** (paid **semi-annually**). The current Market Yield to Maturity (YTM) is **5%**.

Please answer the following questions:

- Q2-1: **Construct a table** listing the **Cash Flow (CF)**, **Discount Factor (DF)**, and **Present Value of Cash Flow (PV of CF)** for each semi-annual period (Year 0.5, 1, 1.5, 2, 2.5, and 3).
- Q2-2: Based on the table above, calculate the weighted average time to maturity, i.e., the **Macaulay Duration**. (Round your answer to 4 decimal places).
- Q2-3: Based on the Macaulay Duration, calculate the bond's **Modified Duration**.
- Q2-4: Explain the **economic meaning** of the "Modified Duration" value you calculated. (i.e., How would the bond price change if the market interest rate shifts slightly?)
- Q2-5: In practice, we can also estimate the Modified Duration by observing price changes corresponding to yield changes. This is known as the **Approximate Modified Duration**.

  Assume the market yield **increases by 10 basis points** (to 5.1%) and **decreases by 10 basis points** (to 4.9%). Calculate the new bond prices for these two scenarios, denoted as P_+ (price when yield rises) and P_- (price when yield falls). Use the following formula to estimate the Modified Duration:

$$\text{Approx. ModDur} = \frac{P_- - P_+}{2 \times P_0 \times \Delta y}$$

  Where P_0 is the current price, and Δy is the change in yield (0.001).

  Compare this result with the exact Modified Duration calculated in Q2-3.

Question 3

Scenario:
Continuing with the data from **Question 2**: Consider the same 3-year Treasury bond with a Face Value of $100 and a Coupon Rate of 6% (paid semi-annually). The current Market Yield to Maturity (YTM) is 5%.
(Note: Please use the current price P_0 and the Modified Duration D_{mod} calculated in Question 2 for the following steps.)

Please answer the following questions:

- Q3-1: Based on the definition of convexity, calculate the exact Convexity value of the bond. Please use the following formula:

$$\text{Convexity} = \frac{1}{P_0 \times m^2 \times (1 + y/m)^2} \sum_{k=1}^{n} \left[\frac{CF_k}{(1 + y/m)^k} \times (k^2 + k) \right]$$

  Where: P_0 is the current bond price, y is the annual YTM (0.05), $m = 2$ is the number of coupon payments per year, CF_k is the cash flow at semi-annual period k, and k is the semi-annual time period.

- Q3-2: In Question 2 (Q2-5), the bond prices were calculated using a yield change of 10 basis points. To obtain a more robust convexity estimate in this question, use a yield change of 100 basis points (BP) ($\Delta y = 0.01$). Calculate P_+ when the yield rises to 6% and P_- when the yield falls to 4%.
  Use these prices to estimate the bond's convexity using the "finite difference method." Please use the following formula:

$$\text{Approx. Convexity} = \frac{P_- + P_+ - 2P_0}{P_0 \times (\Delta y)^2}$$

- Q3-3: Assume the market experiences severe volatility, and the market yield **suddenly rises by 200 basis points** (i.e., $\Delta y = +2\%$ or 0.02).

    1. **Duration Only Model:** Estimate the percentage change in the bond price using only the Modified Duration (D_{mod}) calculated in Question 2.
    Formula: $\frac{\Delta P}{P} \approx -D_{mod} \times \Delta y$
    2. **Duration + Convexity Model:** Estimate the percentage change in the bond price using both the Modified Duration (D_{mod}) and the exact Convexity (C) calculated in Q3-1.
    Formula: $\frac{\Delta P}{P} \approx -D_{mod} \times \Delta y + \frac{1}{2} \times C \times (\Delta y)^2$

- Q3-4: Compare the results from the two models in Q3-3. What is the difference? and Combining with the graph (the convex-to-origin shape of the Price-Yield curve), explain why relying solely on Duration causes significant error when interest rates fluctuate significantly (e.g., 200 bps). How does introducing the "Convexity" term correct this error? (i.e., Is convexity beneficial or detrimental to bondholders?)